```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# --------------------------------
# INPUT IMAGE CREATED INSIDE CODE
# --------------------------------
img = np.zeros((300, 400), dtype=np.uint8)

# Create a sample image
cv2.rectangle(img, (50, 50), (200, 200), 180, -1)
cv2.circle(img, (300, 150), 70, 220, -1)

# Add text
cv2.putText(img, "IMAGE", (90, 270),
            cv2.FONT_HERSHEY_SIMPLEX, 1, 255, 2)

# --------------------------------
# BLUR THE ORIGINAL IMAGE
# --------------------------------
blurred = cv2.GaussianBlur(img, (5, 5), 0)

# --------------------------------
# UNSHARP MASKING
# Formula:
# Sharpened = Original - Blurred
# --------------------------------

unsharp_mask = cv2.subtract(img, blurred)

# Add the unsharp mask to the original image
sharpened = cv2.add(img, unsharp_mask)

# --------------------------------
# DISPLAY RESULTS
# --------------------------------
```

```
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(blurred, cmap='gray')
plt.title("Blurred Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(sharpened, cmap='gray')
plt.title("Unsharp Masking")
plt.axis("off")

plt.show()
```